# Experiment 14

**Question**

**Perform Transformation using Homography Matrix.**

**Aim**

To perform a perspective transformation on an image using a Homography matrix in OpenCV.

**Procedure**

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Select four points from the original image.
6. Select four corresponding points in the output image.
7. Compute the Homography matrix using cv2.getPerspectiveTransform().
8. Apply the transformation using cv2.warpPerspective().
9. Display the transformed image.

**<u>Code</u>**

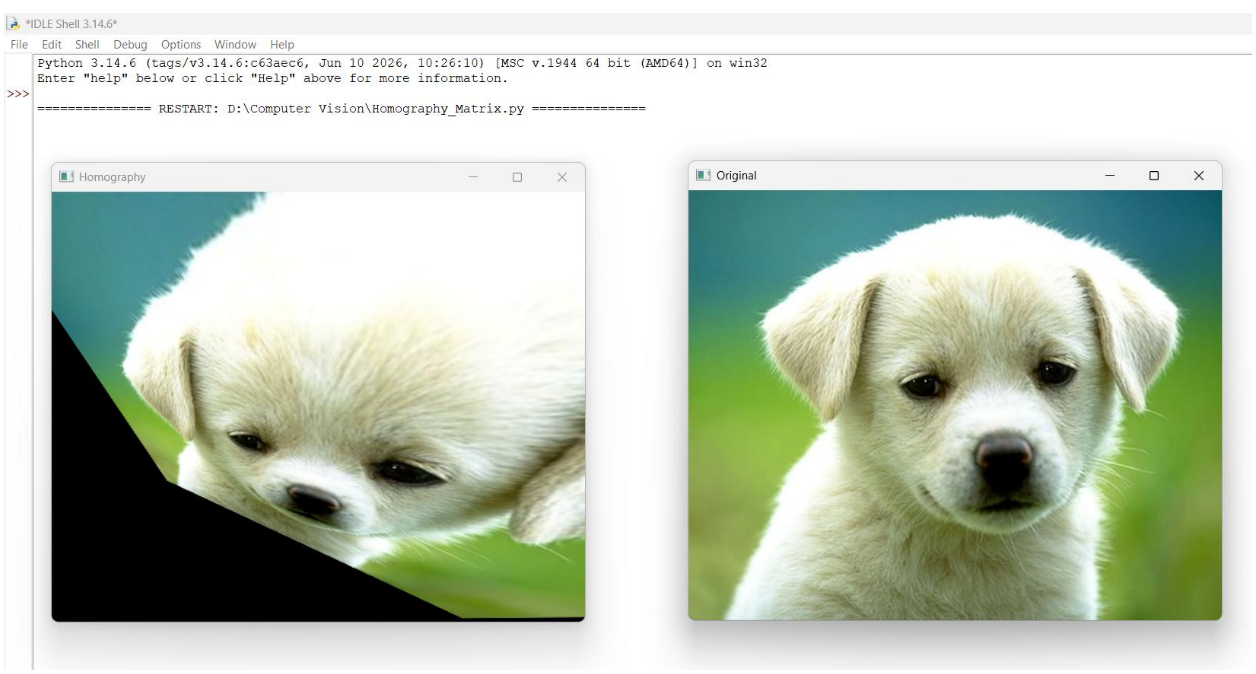
```
import cv2

import numpy as np


img = cv2.imread("dog.jpg")

rows, cols = img.shape[:2]
```

```
pts1 = np.float32([[50,50],[300,50],[50,250],[300,250]])

pts2 = np.float32([[10,100],[300,50],[100,250],[250,300]])


H = cv2.getPerspectiveTransform(pts1, pts2)

result = cv2.warpPerspective(img, H, (cols, rows))


cv2.imshow("Original", img)

cv2.imshow("Homography", result)
```

**Input**

**Input Image:** dog.jpg

**Output :** Homography Transformed Image




**Result**

The Homography transformation was successfully applied to the image using OpenCV. The transformed image was displayed and saved successfully.